

Title

Respiratory muscle training expands thoracic volumes in advanced chronic kidney disease

Abstract

Objective: To determine whether structured inspiratory muscle training reverses restrictive physiology in CKD. **Design:** Randomized controlled trial. **Participants:** 30 adults with CKD stages 3-5 on conservative management (15 progressive-load, 15 fixed-load). **Intervention:** Daily inspiratory training for 8 weeks; progressive group escalated loading to 50% maximal pressure, fixed group maintained 5 cmH₂O. **Outcomes:** Both groups showed significant increases in total thoracic volumes by end of protocol with large effect sizes. The progressive-load group demonstrated significantly greater pulmonary rib cage volume expansion ($F=3.698$, $p=0.042$) compared to fixed-load. Abdominal and chest wall volume changes were similar between groups. Maximal inspiratory pressure increased significantly in the progressive group (73.88 to 90.35 cmH₂O, $p<0.001$) and increased minimally in the fixed group. Maximal expiratory pressure improved in both progressive (97.71 to 108.53 cmH₂O, $p=0.001$) and fixed-load groups (80.69 to 94.77 cmH₂O, $p=0.004$). All participants tolerated training well without adverse events. **Conclusions:** Daily inspiratory muscle training reversed thoracic volume restriction in pre-dialysis CKD, with progressive loading providing greater improvement in inspiratory strength than fixed minimal loading.